

Faculty of Engineering and Technology
B. Tech (CS&E) Programme
Semester VII

Practicals of Natural Language Processing

Course Code: 2604734 **Course Credit: 2**

Programme Objectives (PO):

PO 1 : The graduate will be able to apply the knowledge of mathematics, basic science, engineering fundamentals to solve the engineering problems.

PO 2 : The graduate will be able Identify, formulate and solve computer or IT related problems using acquired knowledge.

PO 3 : The graduate will be able to design software solutions or hardware components as per given specifications.

PO 4 : The graduate will be able to analyse, interpret, conduct investigations using engineering knowledge and provide valid conclusions.

PO 5 : The graduate will be able to use and demonstrate the use of the modern tools and technologies related to the latest trends in the computer industry.

PO 6 : The graduate will be able to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7 : The graduate will be able to understand the impact of the engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8 : The graduate will be able to apply ethical principles and commit to professional ethics and responsibilities.

PO 9 : The graduate will be able to function effectively as an individual, as a member or as a leader in a team.

PO 10 : The graduate will demonstrate competent Communication skills with respect to verbal communication and technical writing like effective reports, documentation and presentations.

PO 11 : The graduate will be able to demonstrate knowledge of Engineering and management principles and will be able to apply to manage projects in multidisciplinary environments.

PO 12 : The graduate will be able to recognize the need for, and will have the ability to engage in self learning as per real world requirements.

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PROGRAM SPECIFIC OUTCOMES (PSOs): **Track-3 (Artificial Intelligence and Machine Learning)**	
At the the end of the program the graduate would be able to:	
PSO 1:	Understand the fundamentals and underlying theoretical and practical concepts of different domains of computer science and engineering.
PSO 2:	Identify and analyze the problem, select the proper platform with appropriate data structure/database, storage and develop the solution with good algorithmic competence and proficiency in a wide range of programming languages.
PSO 3:	Develop and Implement the solutions for real world engineering problems using modern tools and techniques of artificial intelligence and machine learning.

Introduction:

Natural Language Processing (NLP) is a field that enables computers to understand and generate human language. It combines linguistics, computer science, and artificial intelligence. NLP powers applications like chat-bots, machine translation, and sentiment analysis. This course covers key concepts such as tokenization, part-of-speech tagging, and named entity recognition.

Course Objectives:

- To implement different text representation and feature extraction techniques.
- To implement semantic analysis and modeling techniques in NLP.
- To apply advanced deep learning models for NLP applications.
- To apply predictive analytics and deploy NLP models for real-world use cases.

Course Duration: The course duration is 40 sessions of 60 minutes each.

Course Content:

Module No.	Modules/Sub-Modules	No. of Sessions	Marks Weightage (%)
I	Implement text preprocessing techniques, including tokenization, removing stop words, and punctuation, and applying stemming / lemmatization.	**08**	**20%**
II	Implement Bag of Words and TF-IDF for text vectorization and compare the effectiveness of both methods. Demonstrate the N-gram modeling to analyze and establish the probability distribution across sentences and explore the utilization of unigrams, bigrams, and trigrams in diverse English sentences to illustrate the impact of	**08**	**20%**

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	varying n-gram orders on the calculated probabilities. Perform NER and POS tagging on text data to identify entities and their syntactic roles.		
III	Build a text classification model using Word2Vec or GloVe embeddings to classify text data. Perform sentiment analysis on social media text to classify the sentiments.	**08**	**20%**
IV	Using pre-trained models for NLP tasks (BERT, GPT, etc.). Advanced syntactic and semantic analysis using deep learning.	**08**	**20%**
V	Building models for text generation (chatbots) and text summarization. Deploy a text classification model as a REST API that can be used for real-time predictions.	**08**	**20%**

Course Outcomes: At the end of the course the students should be able to:

CO 1: Understand the NLP pipeline, including text preprocessing, tokenization, and vectorization techniques.

CO 2: Apply various text representation methods such as Bag of Words, TF-IDF, and Word2Vec for feature extraction.

CO 3: Analyze semantic relationships in text using lexical semantics, word embeddings, and co-occurrence models.

CO 4: Implement deep learning techniques like RNNs, LSTMs, and Transformers for advanced NLP applications.

CO 5: Evaluate NLP models based on metrics such as accuracy, precision, recall, and deploy them for real-world applications.

Teaching Scheme:

Teaching Scheme (Hours) : 40		
Course Credit: 4		
Theory	Tutorial	Practical
-	-	40

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Examination Scheme:

Examination Scheme				
External	Internal	CEC		Total Marks
		Assignment	Attendance	
50	20	20	10	100

Teaching Methods:

The following pedagogical tools will be used to teach this course:

1. Demonstration
2. Discussions
3. Assignments
4. Presentations
5. Practical Implementations

Supplementary Readings:

1. Jurafsky, D., & Martin, J. H. (2024). Speech and language processing (3rd ed.). Prentice Hall.
2. Geetha, T. V. (2024). Understanding natural language processing: Machine learning and deep learning perspectives (1st ed.). Pearson.
3. Siddiqui, T., & Tiwary, U. S. (2024). Natural language processing and information retrieval. Oxford University Press.
4. Kulkarni, A., & Shivananda, A. (2019). Natural language processing recipes: Unlocking text data with machine learning and deep learning using Python. Apress.
5. Bird, S., Klein, E., & Loper, E. (2009). Natural language processing with Python. O'Reilly Media.

List of Journals / Periodicals / Magazines / Newspapers/ etc.:

The students can refer to the link below to get relevant information pertaining to the subject.

1. https://archive.nptel.ac.in/courses/106/105/106105158/
2. https://archive.nptel.ac.in/courses/106/106/106106211/
3. https://archive.nptel.ac.in/courses/106/101/106101007/

Session Plan:

Session	Topic
1	Implement text preprocessing techniques, including tokenization
2	Implement text preprocessing techniques, including tokenization
3	Implement text preprocessing techniques, removing stop words
4	Implement text preprocessing techniques, removing stop words
5	Implement text preprocessing techniques, removing punctuation
6	Implement text preprocessing techniques, removing punctuation
7	Applying stemming / lemmatization
8	Applying stemming / lemmatization

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9	Implement Bag of Words for text vectorization
10	Implement TF-IDF for text vectorization and compare the effectiveness of both methods
11	Demonstrate the N-gram modeling to analyze and establish the probability distribution across sentences
12	Explore the utilization of unigrams
13	Explore the utilization of bigrams
14	Using trigrams in diverse English sentences to illustrate the impact of varying n-gram orders on the calculated probabilities
15	Perform NER and POS tagging on text data to identify entities and their syntactic roles
16	Perform NER and POS tagging on text data to identify entities and their syntactic roles
17	Build a text classification model using Word2Vec to classify text data
18	Build a text classification model using Word2Vec to classify text data
19	Build a text classification model using Word2Vec to classify text data
20	Build a text classification model using GloVe embeddings to classify text data
21	Build a text classification model using GloVe embeddings to classify text data
22	Perform sentiment analysis on social media text to classify the sentiments
23	Perform sentiment analysis on social media text to classify the sentiments
24	Perform sentiment analysis on social media text to classify the sentiments
25	Using pre-trained models for NLP tasks (BERT, GPT, etc.)
26	Using pre-trained models for NLP tasks (BERT, GPT, etc.)
27	Using pre-trained models for NLP tasks (BERT, GPT, etc.)
28	Using pre-trained models for NLP tasks (BERT, GPT, etc.)
29	Advanced syntactic and semantic analysis using deep learning
30	Advanced syntactic and semantic analysis using deep learning
31	Advanced syntactic and semantic analysis using deep learning
32	Advanced syntactic and semantic analysis using deep learning
33	Building models for text generation (chatbots) and text summarization
34	Building models for text generation (chatbots) and text summarization
35	Building models for text generation (chatbots) and text summarization
36	Building models for text generation (chatbots) and text summarization
37	Deploy a text classification model as a REST API that can be used for real-time predictions
38	Deploy a text classification model as a REST API that can be used for real-time predictions
39	Deploy a text classification model as a REST API that can be used for real-time predictions
40	Deploy a text classification model as a REST API that can be used for real-time predictions

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Suggested Practical Distribution:

The suggested distribution as per Bloom's taxonomy is as follows:

Distribution of practicals for course delivery and evaluation					
Remember	Understand	Apply	Analyse	Evaluate	Create
10%	20%	20%	20%	20%	10%

Instructional Method:

1. The course delivery method will depend upon the requirement of content and the needs of students.
2. The internal evaluation will be done on the basis of a practical exam and a continuous evaluation of students.
3. Students will use supplementary resources such as online videos, NPTEL videos, E courses.

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